

Thatipally Aadithhya

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Education

Indian Institute of Information Technology, Nagpur

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.26/10.0

Nagpur, India

Aug 2023 – Jun 2027

Experience

Futures First

Jul 2026 – Present

Software Development Engineer Intern, Quant & Algo

Hyderabad, India

- Developed low-latency trading pipelines in **C#** and **.NET**, processing **10,000+ trades daily** with millisecond-level execution speed on live production trading workflows.
- Engineered backend order-management services with optimized state-handling logic, reducing order-placement latency from **2s to <10ms** and improving throughput under high-concurrency conditions.
- Implemented **3 risk and compliance controls** (self-trade prevention, price limits, margin checks) and built a **React**-based real-time dashboard for order status and latency monitoring.

AeroAgro AI (IIT Hyderabad)

Apr 2025 – Jul 2025

Project Intern

Hyderabad, India

- Developed a **Node.js** drone-image upload pipeline with input validation, structured error handling, and **Cloudinary** CDN integration, reducing failed transfers by **30%** and enabling image retrieval in under **2 seconds**.
- Collaborated cross-functionally with a **5+ member** engineering and hardware team, translating on-drone requirements into backend features to ensure seamless hardware-software integration.

Projects

Chronicle | C++20, Linux, TCP/UDP, epoll/kqueue, CMake

GitHub

- Built a low-latency electronic trading simulator in **C++20** with a binary **UDP market-data feed**, L2 order-book reconstruction, and a **TCP order-entry gateway** for order acknowledgements and fills.
- Designed deterministic binary **record/replay** and sequence-gap recovery with fault injection for packet loss, duplication, and reordering, enabling reproducible debugging of trading-session failures.
- Implemented a cache-aligned lock-free **SPSC queue** and latency benchmarks for throughput and **p50/p99/p99.9** performance across the event-processing pipeline.

Astra | C, Flex (Lex), Bison (Yacc), GCC

GitHub

- Devised a fully functional compiler from scratch, building a **Flex** lexer for lexical analysis and a **Bison** context-free-grammar parser for syntax parsing of a custom Hindi-inspired programming language.
- Designed an **Abstract Syntax Tree (AST)** and implemented **O(n)** single-pass code generation across **5+ language constructs** (variables, functions, conditionals, loops, I/O) via optimized **DFS traversal**, producing GCC-runable binaries.

LiveDocs - Live Google Docs | TypeScript, Node.js, WebSockets, PostgreSQL

GitHub

- Built a production-grade distributed real-time collaborative document editor supporting **10+ concurrent users** with sub-**50ms** sync latency, implementing a custom **CRDT algorithm** to eliminate merge conflicts across sessions.
- Architected secure multi-user access with **JWT**-authenticated **WebSocket** connections, room-based isolation, and granular per-user permissions across a distributed multi-tiered system.

Technical Skills

Languages: C++, C, C#, Go, Java, JavaScript, TypeScript

Backend & Systems: Linux, TCP/UDP, Multithreading, Distributed Systems, .NET, Node.js, WebSockets

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MySQL, MongoDB, Redis

Developer Tools: Git, GitHub, Linux, CMake, Postman, Vercel

Coursework: System Design, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming

Achievements & Certifications

- **Competitive Programming:** LeetCode Rating **1800+**; solved **400+** problems across data structures and algorithms.
- **Contest Rankings:** Global Rank **769** in CodeChef Contest 234C; solved **200+** problems.
- **Hackathons:** Reached the national semi-finals of Flipkart GRiD 8.0, ranking among the top **1,000** out of **1,65,000+** registrations nationwide.

Leadership Experience

Indian Institute of Information Technology, Nagpur

Feb 2025 – Jul 2025

Student Placement Cell Coordinator

Nagpur, India

- Managed end-to-end placement operations, including interview scheduling, candidate data management, and company coordination, for **50+ companies** and **300+ students**.